DRESS and the WL Hierarchy: Climbing One Deletion at a Time

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Abstract

The Cai–Fürer–Immerman (CFI) construction provides the canonical family of hard instances for the Weisfeiler–Leman (WL) hierarchy: distinguishing the two non-isomorphic CFI graphs over a base graph G requires WL of dimension at least the treewidth of G. In this paper, we introduce Δ^k-DRESS, an iterated extension of the DRESS continuous structural refinement framework that applies k levels of vertex deletion, enumerating all $\binom{n}{k}$ vertex subsets and collecting DRESS fingerprints into a multiset.

We prove that Δ^k-DRESS is at least as powerful as $(k+2)$-WL under the Kelly–Ulam Reconstruction Conjecture: the base case shows that Original-DRESS is at least as powerful as 2-WL via the injectivity of the Birkhoff contraction fixed point, and an inductive argument lifts the result by one WL level per deletion depth, using the fact that the reconstruction deck determines the graph up to isomorphism.

We verify the theoretical bound empirically on the canonical CFI benchmark: Original-DRESS ($k = 0$) already distinguishes CFI(K_3) (requiring 2-WL), and each additional level extends the range by one WL level, up to $k = 3$ matching 5-WL. Crucially, $k = 3$ fails on CFI(K_7) (requiring 6-WL), confirming a sharp boundary at $(k+2)$-WL. The computational cost is $\mathcal{O}\big(\binom{n}{k} \cdot I \cdot m \cdot d_{\max}\big)$—polynomial in n for fixed k. These results establish iterated DRESS as a practical framework for systematically climbing the WL hierarchy.

1 Introduction

The Weisfeiler–Leman (WL) hierarchy [10] is the standard yardstick for measuring the expressiveness of graph isomorphism algorithms. The j-WL test operates on j-tuples of vertices and runs in $\mathcal{O}(n^{j+1})$ time per iteration [2]. While powerful in theory, even 3-WL at $\mathcal{O}(n^4)$ is impractical for large graphs.

The CFI construction [2] provides the canonical hard instances: for any base graph G, it produces a pair of non-isomorphic graphs CFI(G) and CFI$'(G)$ that j-WL cannot distinguish whenever j is below the treewidth of G. In particular, CFI(K_n) requires $(n-1)$-WL, making these the hardest known instances for the WL hierarchy.

In the companion paper [4], we introduced the DRESS family of continuous structural refinement algorithms. Δ-DRESS, which runs DRESS on each vertex-deleted subgraph, was shown to empirically distinguish strongly regular graphs that confound 3-WL and connects to the Kelly–Ulam reconstruction conjecture [8, 9].

In this paper, we introduce Δ^k-DRESS, which applies k levels of vertex *deletion*. Under the Kelly–Ulam Reconstruction Conjecture [8, 9]—which asserts that a graph with $n \geq 3$ vertices is determined up to isomorphism by its deck of vertex-deleted subgraphs—deletion is equivalent to individualization in discriminative power. Since $(j+1)$-WL is equivalent to running j-WL over all single-vertex individualizations [2, 6], the reconstruction deck provides precisely the information needed to climb the WL hierarchy.

We verify Δ^k-DRESS on the CFI family: $k = 0$ (Original-DRESS) already distinguishes CFI(K_3), $k = 1$ reaches CFI(K_4), $k = 2$ reaches CFI(K_5), and $k = 3$ reaches CFI(K_6).

1.1 Contributions

1. We define Δ^k-DRESS (iterated vertex deletion) formally and analyze its complexity.
2. We prove that Δ^k-DRESS is at least as powerful as $(k+2)$-WL under the Kelly–Ulam Reconstruction Conjecture: Original-DRESS matches 2-WL via injectivity of the Birkhoff contraction fixed point, and each deletion level adds one WL level by induction (Theorems 5 and 8).
3. We verify the theoretical bound empirically on the canonical CFI benchmark family, showing that Δ^k-DRESS distinguishes $\mathrm{CFI}(K_{k+3})$ for each $k = 0, 1, 2, 3$, and fails at the next level, covering WL levels 2 through 5.
4. We observe a sharp boundary at $(k+2)$-WL: $k = 3$ fails on $\mathrm{CFI}(K_7)$ (requiring 6-WL).
5. We discuss the theoretical implications and open questions raised by these results.

2 Preliminaries

2.1 DRESS

The Original-DRESS equation [3] is a parameter-free, non-linear dynamical system on edges:

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} \left(d_{ux}^{(t)} + d_{xv}^{(t)} \right)}{\|u\|^{(t)} \cdot \|v\|^{(t)}} \tag{1}$$

where $\|u\|^{(t)} = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$ is the vertex norm and $N[u] = N(u) \cup \{u\}$ is the closed neighborhood. Self-loops are added to every vertex; the invariant $d_{uu} = 2$ holds at every iteration. The equation converges to a unique fixed point $d^* \in [0, 2]^{|E|}$ for any positive initialization [4].

The graph fingerprint can be represented equivalently as the sorted vector $\mathrm{sort}(d^*)$ or as the histogram $h(d^*)$; both uniquely identify the multiset of converged edge values. Since DRESS values are bounded in $[0, 2]$ and convergence tolerance is $\epsilon = 10^{-6}$, each edge value maps to one of $\lceil 2/\epsilon \rceil = 2 \times 10^6$ integer bins (the bin count is determined by the convergence tolerance), so the histogram is a compact bin-count vector. In the experiments we use the histogram representation for efficiency; in the proofs we work with the full multiset fingerprint. Two graphs are declared non-isomorphic if their fingerprints differ.

2.2 Δ-DRESS

Δ-DRESS [4] runs DRESS on each vertex-deleted subgraph:

Definition 1 (Δ-DRESS)**.** *Given a graph $G = (V, E)$ and a DRESS variant $\mathcal{F}$, the Δ-DRESS fingerprint is the histogram obtained by running $\mathcal{F}$ on each vertex-deleted subgraph and accumulating every edge similarity value into a single bin-count vector:*

$$\Delta\text{-}DRESS(\mathcal{F}, G) = h\left(\bigsqcup_{v \in V} \mathcal{F}(G \setminus \{v\}) \right)$$

where $G \setminus \{v\}$ is the subgraph induced by $V \setminus \{v\}$, $\bigsqcup$ denotes multiset union, and h maps each value to its integer bin in a fixed-size array.

3 Δ^k-DRESS

We now define the iterated vertex deletion operator.

Definition 2 (Δ^k-DRESS — Deletion)**.** *For a graph $G = (V, E)$, a DRESS variant $\mathcal{F}$, and a deletion depth $k \geq 0$:*

$$\Delta^k\text{-}DRESS(\mathcal{F}, G) = h\left(\bigsqcup_{S \subset V,\, |S|=k} \mathcal{F}(G \setminus S)\right)$$

where $G \setminus S$ is the subgraph induced by $V \setminus S$, and h maps the pooled edge values into a histogram.

Proposition 3 (Complexity)**.** *The time complexity of Δ^k-DRESS is*

$$\mathcal{O}\left(\binom{n}{k} \cdot I \cdot m \cdot d_{\max}\right)$$

where $n = |V|$, $m = |E|$, $d_{\max}$ is the maximum degree, and I is the number of DRESS iterations per subgraph. The $\binom{n}{k}$ computations are entirely independent and embarrassingly parallel.

Comparison with j-WL. The cost of j-WL is $\mathcal{O}(n^{j+1})$ per iteration. For CFI(K_6), 5-WL costs $\mathcal{O}(n^6)$, while Δ^3-DRESS costs $\mathcal{O}(n^3 \cdot I \cdot m \cdot d_{\max}) = \mathcal{O}(n^3 \cdot m \cdot d_{\max})$ (with I constant).

4 Expressiveness Relative to the Weisfeiler–Lehman Hierarchy

We now prove that Δ^k-DRESS is at least as powerful as the $(k+2)$-dimensional Weisfeiler–Lehman ($(k+2)$-WL) test under the Kelly–Ulam Reconstruction Conjecture. We first establish the base case, then lift the result by induction on the deletion depth.

We first establish a key property of the DRESS fixed-point operator.

Lemma 4 (Refinement monotonicity)**.** *Let F denote the Original-DRESS update operator, and let $\mathbf{d}^{(t)}$ denote the edge-value vector at iteration t. If two edges (u, v) and (u', v') have the same value $d_{uv}^{(t)} = d_{u'v'}^{(t)}$ at iteration t but differ in their common-neighbor multiset structure—i.e., the multisets $\{\{(d_{ux}^{(t)}, d_{xv}^{(t)}) : x \in N[u] \cap N[v]\}\}$ and $\{\{(d_{u'x'}^{(t)}, d_{x'v'}^{(t)}) : x' \in N[u'] \cap N[v']\}\}$ are not equal—then $d_{uv}^{(t+1)} \neq d_{u'v'}^{(t+1)}$.*

Proof. The DRESS update for edge (u, v) is:

$$d_{uv}^{(t+1)} = \frac{\sum_{x \in N[u] \cap N[v]} (d_{ux}^{(t)} + d_{xv}^{(t)})}{\|u\|^{(t)} \cdot \|v\|^{(t)}}$$

where $\|u\|^{(t)} = \sqrt{\sum_{x \in N[u]} d_{ux}^{(t)}}$. This is a rational function of the input values. At each iteration, the numerator is a sum over the common-neighbor multiset $N[u] \cap N[v]$, and the denominator depends on the full closed neighborhoods $N[u]$ and $N[v]$.

Suppose, for contradiction, that two edges with distinct common-neighbor multiset structures always produce the same output. This would require the rational equation

$$\frac{\sum_{x \in N[u] \cap N[v]} (d_{ux} + d_{xv})}{\|u\| \cdot \|v\|} = \frac{\sum_{x' \in N[u'] \cap N[v']} (d_{u'x'} + d_{x'v'})}{\|u'\| \cdot \|v'\|}$$

to hold identically for all positive $\mathbf{d}$, despite the two sides being structurally different rational functions (aggregating over different-sized or differently-weighted multisets). Since distinct multiset structures yield distinct polynomial expressions in the numerator and/or denominator, equality cannot hold on an open set of positive vectors. In particular, the unique fixed point $\mathbf{d}^*$ (which lies in the interior of the positive cone by the Birkhoff contraction theorem [4]) is a generic point of this open set, so the two values must differ at $\mathbf{d}^*$. □

Theorem 5 (Original-DRESS is at least as powerful as 2-WL)**.** *Let G and H be two graphs. If the stable colorings produced by the 2-dimensional Weisfeiler–Lehman test on G and H differ, then the DRESS fingerprints of G and H differ (i.e.,* $\mathrm{sort}(\mathbf{d}^*_G) \neq \mathrm{sort}(\mathbf{d}^*_H)$ *or* $h(\mathbf{d}^*_G) \neq h(\mathbf{d}^*_H)$*).*

Proof. The stable 2-WL assigns the same color to two pairs (u, v) and (u', v') if and only if they have identical multisets of common-neighbor colors.

Consider the fixed-point equation of Original-DRESS (triangle motif):

$$d^*_{uv} \cdot \|u\| \cdot \|v\| = \sum_{x \in N[u] \cap N[v]} (d^*_{ux} + d^*_{xv}). \tag{2}$$

The right-hand side aggregates over precisely the same common-neighbor multiset that 2-WL uses for refinement.

We show that DRESS refines at least as finely as 2-WL. The 2-WL coloring is computed by iterated multiset refinement: at each round, pairs with distinct common-neighbor color multisets receive distinct colors. DRESS performs an analogous iteration in continuous space: at each step, the value $d^{(t+1)}_{uv}$ is a function of the multiset $\{\{(d^{(t)}_{ux}, d^{(t)}_{xv}) : x \in N[u] \cap N[v]\}\}$. By Lemma 4, whenever two edges have distinct common-neighbor value-multisets, they receive distinct updated values. Since DRESS starts from a uniform initialization (all edges equal) and 2-WL starts from a uniform coloring (all pairs same color), an easy induction on the iteration count shows that:

If 2-WL assigns different colors to (u, v) and (u', v') at round t, then $d^{(t)}_{uv} \neq d^{(t)}_{u'v'}$.

At the stable coloring (the fixed point of both processes), any color difference in 2-WL is therefore reflected as a value difference in $\mathbf{d}^*$. In particular, if the 2-WL stable colorings of G and H differ—meaning the multisets of stable colors are not equal—then the multisets of fixed-point edge values $\mathbf{d}^*_G$ and $\mathbf{d}^*_H$ are not equal either, and hence $\mathrm{sort}(\mathbf{d}^*_G) \neq \mathrm{sort}(\mathbf{d}^*_H)$. □

Before stating the main inductive theorem, we recall a classical characterization of the WL hierarchy via vertex individualization.

Lemma 6 (WL and vertex individualization)**.** *Let $j \geq 2$ and let G, H be two graphs. If $(j+1)$-WL distinguishes G from H, then the multisets of j-WL colorings over all single-vertex individualizations differ:*

$$\{\{j\text{-}WL(G, v) : v \in V(G)\}\} \neq \{\{j\text{-}WL(H, w) : w \in V(H)\}\}$$

where (G, v) denotes the graph G with vertex v individualized (given a unique color).

Proof. This is a consequence of the characterization of j-WL via the counting logic C^j [2, 7]. The logic C^{j+1} can express "there exists a vertex v such that $\varphi(v)$ holds," where φ is a C^j formula. Since $(j+1)$-WL captures exactly C^{j+1} [2], any distinction made by $(j+1)$-WL can be witnessed by individualizing one vertex and applying j-WL. This is formalized in [6, Theorem 5.1]; see also [5, Section 2] for a self-contained treatment. □

We now define the **multiset fingerprint** of Δ^k-DRESS, which retains full per-deletion information:

Definition 7 (Multiset fingerprint — Deletion)**.** *For a graph $G = (V, E)$, a DRESS variant $\mathcal{F}$, and a deletion depth $k \geq 0$:*

$$\Delta^k\text{-}DRESS_{\mathrm{ms}}(\mathcal{F}, G) = \{\{\mathrm{sort}(\mathcal{F}(G \setminus S)) : S \subset V,\ |S| = k\}\}$$

i.e., the multiset of per-deletion sorted fingerprint vectors. This is strictly more informative than the pooled histogram of Definition 2.

Theorem 8 (Δ^k-DRESS is at least as powerful as $(k+2)$-WL under RC)**.** *Assume the Kelly–Ulam Reconstruction Conjecture holds. For any fixed integer $k \geq 0$, if the $(k+2)$-dimensional Weisfeiler–Lehman test distinguishes two graphs G and H, then their Δ^k-DRESS multiset fingerprints differ:*

$$\Delta^k\text{-}DRESS_{\text{ms}}(\mathcal{F}, G) \neq \Delta^k\text{-}DRESS_{\text{ms}}(\mathcal{F}, H).$$

Proof. We proceed by induction on k, assuming the Reconstruction Conjecture throughout.

Base case ($k = 0$): Δ^0-DRESS reduces to Original-DRESS (no vertices are deleted, and the empty subset $S = \emptyset$ contributes one run). The multiset fingerprint is $\{\{\text{sort}(\mathbf{d}^*_G)\}\}$. The claim is exactly Theorem 5.

Inductive hypothesis: Assume the statement holds for depth $k-1$: whenever $(k+1)$-WL distinguishes two graphs, their Δ^{k-1}-DRESS multiset fingerprints differ (under RC).

Inductive step: Let $(k+2)$-WL distinguish G and H. By Lemma 6, the multisets of $(k+1)$-WL colorings over all single-vertex individualizations differ:

$$\{\{(k{+}1)\text{-WL}(G, v) : v \in V(G)\}\} \neq \{\{(k{+}1)\text{-WL}(H, w) : w \in V(H)\}\}.$$

Under the Reconstruction Conjecture, the deck of vertex-deleted subgraphs $\{G \setminus \{v\} : v \in V(G)\}$ determines G up to isomorphism. In particular, any structural distinction that $(k{+}1)$-WL detects via vertex individualization is also detectable from the vertex-deleted subgraphs: if $(k{+}1)$-WL assigns different colorings to (G, v) and (H, w), then the subgraphs $G \setminus \{v\}$ and $H \setminus \{w\}$ are structurally distinguishable.

Now consider the Δ^k-DRESS multiset fingerprint. Every subset $S \subset V(G)$ with $|S| = k$ can be decomposed as $S = \{v\} \cup S'$ where $|S'| = k-1$. Since $G \setminus S = (G \setminus \{v\}) \setminus S'$, collecting over all $(k{-}1)$-subsets S' for a fixed v yields precisely Δ^{k-1}-DRESS applied to $G \setminus \{v\}$:

$$\Delta^k\text{-DRESS}_{\text{ms}}(\mathcal{F}, G) = \biguplus_{v \in V(G)} \Delta^{k-1}\text{-DRESS}_{\text{ms}}(\mathcal{F}, G \setminus \{v\}).$$

Since the vertex-deleted subgraphs $G \setminus \{v\}$ and $H \setminus \{w\}$ are structurally distinguishable (under RC) whenever the corresponding individualizations are, and $(k{+}1)$-WL distinguishes the relevant pairs, the inductive hypothesis guarantees that Δ^{k-1}-DRESS on $G \setminus \{v\}$ differs from Δ^{k-1}-DRESS on $H \setminus \{w\}$. Since the multisets of $(k{+}1)$-WL individualizations differ between G and H, the multisets of Δ^{k-1}-DRESS fingerprints over all single-vertex deletions also differ. This difference is preserved in the multiset union, completing the induction. □

Remark 9. *The proof is stated for the triangle-based Original-DRESS update. It extends verbatim to any Motif-DRESS variant (with the base case becoming $\geq (r+1)$-WL when the motif M has size $r+1$). The deletion mechanism that climbs the hierarchy remains unchanged.*

Remark 10 (Histogram representation)**.** *Δ^k-DRESS uses pooled histograms in practice. The multiset fingerprint (which retains per-subset identity) is strictly more informative and provides the guarantee. The pooled histogram is a coarsening: different multisets could in principle produce the same histogram. Nevertheless, the pooled histogram suffices empirically for all CFI instances tested (Table 1).*

5 Empirical Results

All experiments use Original-DRESS as the base variant $\mathcal{F}$, with convergence tolerance $\epsilon = 10^{-6}$ and a maximum of 100 iterations. In every case, convergence was reached before the iteration limit.

5.1 CFI Graphs

The CFI construction [2] is defined over a base graph $G = (V_G, E_G)$. For each vertex $v \in V_G$, a "gadget" replaces v with a set of vertices encoding its incident edges. Two non-isomorphic graphs CFI(G) and CFI$'(G)$ are produced by twisting the parity of one gadget. The key property is:

Theorem 11 (Cai, Fürer, Immerman [2])**.** *The j-WL algorithm cannot distinguish CFI(G) from CFI$'(G)$ whenever $j < tw(G)$, where $tw(G)$ is the treewidth of G.*

For the complete graph K_n, the treewidth is $n-1$, so CFI(K_n) requires $(n-1)$-WL. In particular:

- CFI(K_3): requires 2-WL.
- CFI(K_5): requires 4-WL, operating on 4-tuples—$\mathcal{O}(n^5)$ per iteration.
- CFI(K_6): requires 5-WL, operating on 5-tuples—$\mathcal{O}(n^6)$ per iteration.

5.2 Δ^k-DRESS on CFI(K_n)

We test Δ^k-DRESS for $k = 0, 1, 2, 3$ on CFI pairs over complete graphs K_n ($n = 3, \ldots, 10$). Higher depths on larger base graphs were not executed due to time constraints (marked — in the table).

Table 1: Δ^k-DRESS results on CFI graph pairs. ✓ = pair distinguished; × = pair not distinguished; — = not executed due to time constraints. The WL requirement column shows the minimum WL dimension needed to distinguish the pair.

Base graph	$\lvert V_{\text{CFI}}\rvert$	WL req.	Δ^0	Δ^1	Δ^2	Δ^3
K_3	6	2-WL	✓	✓	✓	✓
K_4	16	3-WL	×	✓	✓	✓
K_5	40	4-WL	×	×	✓	✓
K_6	96	5-WL	×	×	×	✓
K_7	224	6-WL	×	×	×	×
K_8	512	7-WL	×	×	—	—
K_9	1152	8-WL	×	×	—	—
K_{10}	2560	9-WL	×	×	—	—

5.3 Analysis

The results in Table 1 reveal a clear staircase pattern:

1. Δ^0 **(Original-DRESS)** distinguishes CFI(K_3), which requires 2-WL. This is a notable result: Original-DRESS—with no vertex deletion—already exceeds 1-WL on CFI instances, consistent with the prism vs. $K_{3,3}$ result in [4]. It fails on all larger CFI pairs.
2. Δ^1 extends the range to CFI(K_4) (3-WL). This is consistent with the results in [4], where Δ-DRESS was shown to distinguish SRG pairs that confound 3-WL. It fails on CFI(K_5).
3. Δ^2 reaches CFI(K_5) (4-WL).
4. Δ^3 reaches CFI(K_6) (5-WL) but fails on CFI(K_7) (6-WL), confirming that the staircase has a sharp boundary at $(k+2)$-WL. The CFI(K_7) test required processing all $\binom{224}{3}$ = 1,848,224 subgraphs (≈19.3 billion edge values).

WL level per deletion depth. The tested data shows:

Deletion depth k	Max WL matched	Effective WL
0	2-WL	$k+2$
1	3-WL	$k+2$
2	4-WL	$k+2$
3	5-WL	$k+2$

The pattern is strikingly regular: Δ^k-DRESS empirically matches $(k+2)$-WL on the CFI family, consistent with the lower bound proved in Theorem 8. Each level adds one WL level of expressiveness, with the base DRESS contributing 2-WL on its own. The computational cost grows as $\binom{n}{k}$ (polynomial in n for fixed k), while the equivalent $(k+2)$-WL cost grows as n^{k+3}.

6 Discussion

6.1 Relationship to Existing Work

Subgraph GNNs. Methods such as ESAN [1] and GNN-AK+ [11] also use vertex-deleted or vertex-marked subgraphs to boost expressiveness. However, these are supervised methods that learn aggregation functions from data. Δ^k-DRESS is entirely unsupervised: the aggregation is the deterministic DRESS fixed point, and the histogram comparison is parameter-free. Δ^k-DRESS corresponds directly to the vertex-deletion approach used in subgraph GNNs.

Reconstruction conjecture. The Kelly–Ulam conjecture [8, 9] states that graphs with $n \geq 3$ are determined by their deck of vertex-deleted subgraphs. Conceptually, Δ^1-DRESS computes a continuous relaxation of this deck: each "card" is the DRESS fingerprint of a vertex-deleted subgraph. The expressiveness proof (Theorem 8) relies on the Reconstruction Conjecture to bridge vertex deletion and the WL individualization hierarchy. The empirical success of Δ^k-DRESS on all tested CFI instances is consistent with RC, and provides further evidence for the conjecture on these graph families.

Fixed-parameter tractability. For fixed k, Δ^k-DRESS runs in polynomial time. Theorem 8 guarantees (under RC) that Δ^k-DRESS is at least as powerful as $(k+2)$-WL, and the CFI experiments confirm this bound is tight: the minimum depth needed to distinguish $\mathrm{CFI}(K_n)$ grows as $n-2$, which is linear in n. While each additional level is polynomial, the number of required levels grows with the base graph size.

6.2 Open Questions

1. **Does the staircase pattern continue beyond $k = 3$?** Theorem 8 establishes $\geq (k+2)$-WL (under RC). We have confirmed empirically that $k = 3$ fails on $\mathrm{CFI}(K_7)$ (6-WL), so the bound is tight for $k \leq 3$. Testing $k = 4$ on $\mathrm{CFI}(K_7)$ would confirm whether the staircase extends.

2. **Is the tightness specific to CFI, or does it hold more broadly?** The lower bound $(k+2)$-WL is general, but the upper bound may depend on the graph family.

3. **Are there graph families where Δ^k-DRESS fails for all constant k?** CFI graphs over graphs of unbounded treewidth would be natural candidates.

4. **Does combining iterated deletion with other DRESS variants yield further gains?** The DRESS family includes Motif-DRESS and Generalized-DRESS [4]. Whether

a stronger base variant $\mathcal{F}$ shifts the staircase upward or merely overlaps with the deletion mechanism is an open question.

5. **Can the RC assumption be removed?** Theorem 8 relies on the Reconstruction Conjecture. An interesting direction is whether the expressiveness guarantee can be proved unconditionally, perhaps via alternative structural arguments.

7 Conclusion

We introduced Δ^k-DRESS, which applies k levels of iterated vertex deletion to the DRESS framework. We proved that Δ^k-DRESS is at least as powerful as $(k+2)$-WL under the Kelly–Ulam Reconstruction Conjecture (Theorem 8): the reconstruction deck bridges vertex deletion and the WL individualization hierarchy, allowing the inductive argument to lift one WL level per deletion depth. Δ^k-DRESS was verified empirically on the canonical CFI benchmark family: at $k = 0$, Original-DRESS already distinguishes CFI(K_3), and each additional level adds one WL level. $k = 3$ reaches 5-WL but fails on 6-WL, confirming the bound is tight on CFI instances. These results establish iterated DRESS as a systematic, scalable method for climbing the WL hierarchy with provable expressiveness guarantees (modulo RC). An open-source implementation is available at `https://github.com/velicast/dress-graph`.

References

[1] Beatrice Bevilacqua, Fabrizio Frasca, Derek Lim, Balasubramaniam Srinivasan, Chen Cai, Gopinath Balamurugan, Michael M. Bronstein, and Haggai Maron. Equivariant subgraph aggregation networks. In *International Conference on Learning Representations (ICLR)*, 2022.

[2] Jin-Yi Cai, Martin Fürer, and Neil Immerman. An optimal lower bound on the number of variables for graph identification. *Combinatorica*, 12(4):389–410, 1992.

[3] Eduar Castrillo, Elizabeth León, and Jonatan Gómez. Dynamic structural similarity on graphs. *arXiv preprint arXiv:1805.01419*, 2018.

[4] Eduar Castrillo Velilla. DRESS: A continuous framework for structural graph refinement, 2026. arXiv preprint arXiv:2602.20833.

[5] Holger Dell, Martin Grohe, and Gaurav Rattan. Lovász meets Weisfeiler and Leman. *Proceedings of the 45th International Colloquium on Automata, Languages, and Programming (ICALP)*, pages 40:1–40:14, 2018.

[6] Martin Grohe. *Descriptive Complexity, Canonisation, and Definable Graph Structure Theory*, volume 47 of *Lecture Notes in Logic*. Cambridge University Press, 2017.

[7] Neil Immerman and Eric Lander. Describing graphs: A first-order approach to graph canonization. *Complexity Theory Retrospective*, pages 59–81, 1990.

[8] Paul J. Kelly. *On Isometric Transformations*. PhD thesis, University of Wisconsin, 1942.

[9] Stanislaw M. Ulam. *A Collection of Mathematical Problems*. Interscience Publishers, 1960.

[10] Boris Weisfeiler and Andrei Leman. The reduction of a graph to canonical form and the algebra which appears therein. *Nauchno-Technicheskaya Informatsia*, 2(9):12–16, 1968.

[11] Lingxiao Zhao, Wei Jin, Leman Akoglu, and Neil Shah. From stars to subgraphs: Uplifting any GNN with local structure awareness. In *International Conference on Learning Representations (ICLR)*, 2022.